SCSB2501 COMPUTER NETWOKS LAB

SAMPLE PROGRAM 1:

Creating a Simple LAN Network using Cisco packet tracer (Network Simulation Tool)

ALGORITHM

Step 1. Open Cisco Packet Tracer

Step 2. Drag and Drop Devices

- Go to **End Devices** (bottom left panel)
  → Drag **2 PCs** onto the workspace.
- Go to **Switches**
  → Drag **1 Switch (e.g., 2960)** onto the workspace.

Step 3. Connect Devices using Copper Straight-Through Cable

- Select **Connections (lightning icon)** → Choose **Copper Straight-Through**
- Connect:
  - **PC0 → FastEthernet0 → Switch → FastEthernet0/1**
  - **PC1 → FastEthernet0 → Switch → FastEthernet0/2**

Step 4. Assign IP Addresses

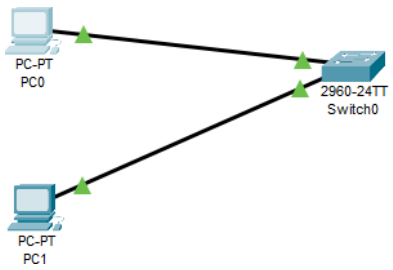
- Click on **PC0** → Desktop → IP Configuration
  - IP Address: 192.168.1.1
  - Subnet Mask: 255.255.255.0
- Click on **PC1** → Desktop → IP Configuration
  - IP Address: 192.168.1.2
  - Subnet Mask: 255.255.255.0

Step 5. Test the Network

- Click on **PC0** → Desktop → Command Prompt

Type: ping 192.168.1.2

OUTPUT:



1 a. To find Network IP address of a machine, time and date server.

Step:1 Place and Connect Devices

- Drag a **Switch**, **PC**, **Server and Router** onto the workspace.
- Connect with **Copper Straight-Through cables**:
    - PC → Switch
    - Server → Switch
    - Switch-> Router

Step2: Assign IP Addresses

Click on PC0 > Desktop > IP Configuration:

- **IP Address:** 192.168.1.1
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.1.1 (optional for LAN)

Click on Server > Config:

- **FastEthernet IP Address:** 192.168.1.100
- Enable the following services (on left pane):
    - **NTP** – Set current time.
    - **FTP / HTTP** – Add some files or enable default site.

step3: Enable Services on Server

- Go to **Services tab** on Server.
- Under **NTP**, make sure it's **ON** and set current time.
- Under **FTP/HTTP**, ensure **FTP/HTTP is ON** and you can add sample files if needed.

Step4: Router Configuration -> CLI

- Assign Router IP Address**:** 192.168.1.1.3
- Assign Router IP address Gateway to PC and Server
- Configure router using CLI commands

>enable

>configure terminal

>interface GigabitEthernet0/0

>ip address 192.168.1.1 255.255.255.0

>no shutdown

>exit

Step5: To Show IP Address, Time And Date of the Server Machine in Router

>enable

>configure terminal

>ntp server 192.168.1.100

>end

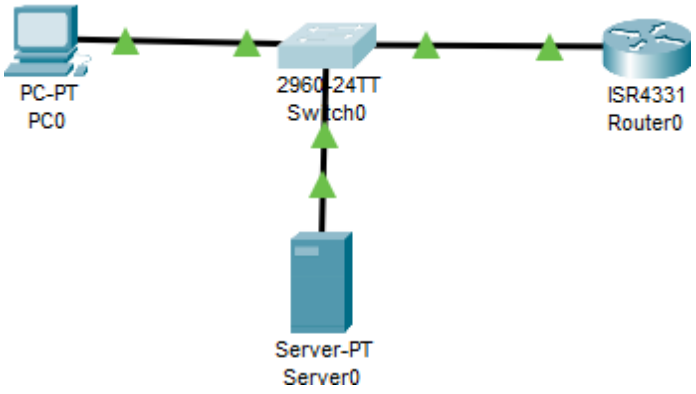
>show ntp status

>show clock

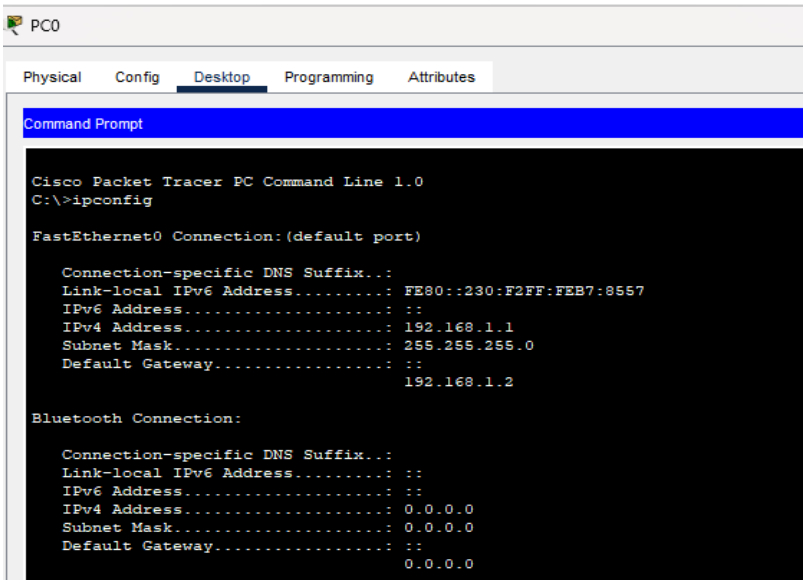
OUTPUT- 1 Date and Time

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ntp server 192.168.1.100
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ntp status
Clock is synchronized, stratum 2, reference is 192.168.1.100
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is EBDB5DFB.0000028A (20:50:3.650 UTC Tue Jun 24 2025)
clock offset is 0.00 msec, root delay is 0.00  msec
root dispersion is 19.36 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system
poll interval is 4, last update was 9 sec ago.
Router#
% Unknown command or computer name, or unable to find computer address

Router#show clock
20:50:34.362 UTC Tue Jun 24 2025
```



Output 2: IP Address of the machine(pc)
Goto: Pc-> desktop->command prompt
>ipconfig (Command for IP address)



1.B. Working Complex Topology

Creating a **hybrid network topology** in **Cisco Packet Tracer** involves combining two or more standard topologies—such as **star, bus, ring, mesh, and tree.**

Hybrid Topology: Star + Bus

Devices:

- 2 Routers (for backbone)
- 2 Switches (for star segments)
- 6 PCs (3 per switch)

Assign IP Addresses for Routers

- **Router0 GigabitEthernet0/0**: 192.168.1.1
- **Router0 GigabitEthernet0/0/1**: 10.0.0.1
- **Router1 GigabitEthernet0/0**: 192.168.2.1
- **Router1 GigabitEthernet0/0/1**: 10.0.0.2

PCs for Switch 1 (pc-> IP configure-> set IP and Default Gateway)

- PC0-> IP: 192.168.1.2 / Default Gateway: 192.168.1.1
- PC1-> IP: 192.168.1.3 / Default Gateway: 192.168.1.1
- PC2-> IP: 192.168.1.4 / Default Gateway: 192.168.1.1

PCs Switch 2(pc-> IP configure-> set IP and Default Gateway)

- PC3-> IP: 192.168.2.2/ Default Gateway: 192.168.2.1
- PC4-> IP: 192.168.2.3/ Default Gateway: 192.168.2.1
- PC5-> IP: 192.168.2.4/ Default Gateway: 192.168.2.1

Configure Routers

Use the **CLI** on each router:

Router0

enable
configure terminal
interface **GigabitEthernet**0/0/0
ip address 192.168.1.1 255.255.255.0
no shutdown

interface **GigabitEthernet**0/0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
end
configure terminal
ip route 192.168.2.0 255.255.255.0 10.0.0.2 (connecting to R1)

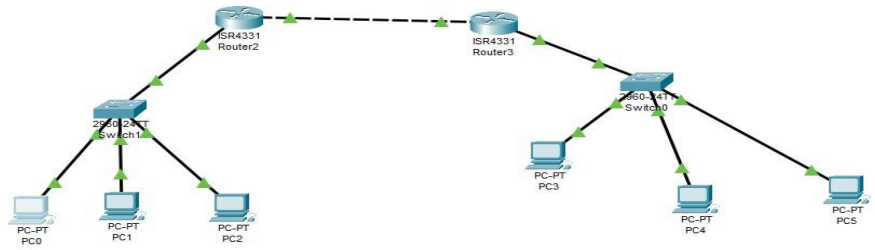
Router1

enable
configure terminal
interface **GigabitEthernet**0/0/0
ip address 192.168.2.1 255.255.255.0
no shutdown

interface **GigabitEthernet**0/0/0
ip address 10.0.0.2 255.255.255.252
no shutdown
end
configure terminal
ip route 192.168.1.0 255.255.255.0 10.0.0.1 (connecting to R0)

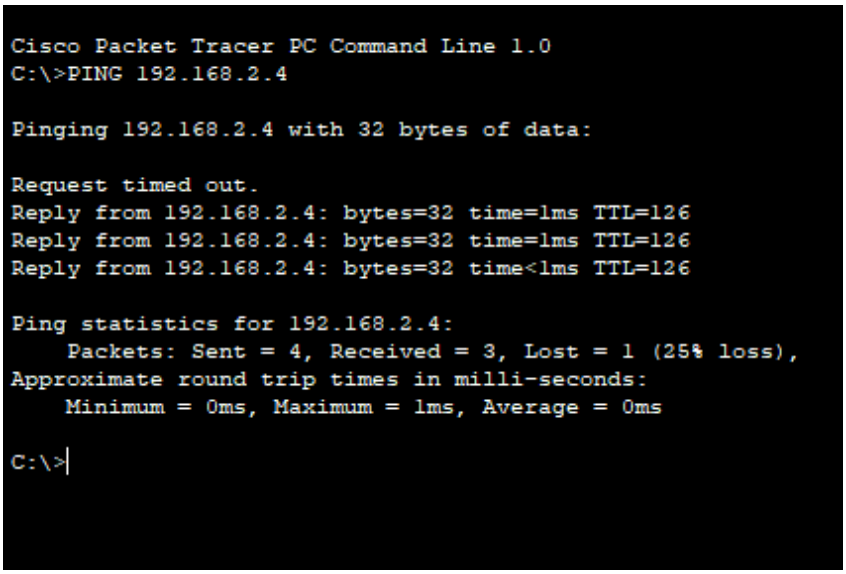
6. **Test Connectivity**

Output:



Use the **ping** tool from one PC to another across the routers to test end-to-end connectivity.

Eg. Ping 192.168.2.4 (from pc0)



1C. To monitor network traffic using Wire Shark step by step

ALGORITHM

Step 1: Launch Wireshark

- Open **Wireshark** as Administrator (on Windows) or with sudo (on Linux) for full access to interfaces.

You will see a list of available **network interfaces** (e.g., Wi-Fi, Ethernet, Loopback).

Step 2: Choose the Interface to Monitor

- Identify and **select your active network interface** (e.g., Wi-Fi if you're using a wireless connection).
- You can recognize active interfaces by **the real-time graph showing live traffic**.
- Double-click the interface to start capturing traffic.

Step 3: Start Capturing Packets

- **packet capture begins**.
- You'll see a live stream of packets showing up, each with:
    - **Time**
    - **Source IP**
    - **Destination IP**
    - **Protocol (e.g., TCP, UDP, DNS, HTTP)**
    - **Info (brief detail)**

Step 4: Apply Filters (Optional but Recommended)

- Wireshark captures **all packets**, which can be overwhelming.
- Apply a **display filter** to narrow down what you see.
    - Examples:
        - http → Show only HTTP traffic
        - ip.addr == 192.168.0.1 → Show traffic to/from that IP
        - tcp.port == 80 → Show HTTP (port 80) traffic
        - dns → Show only DNS queries

Filters go in the **top filter bar** (just below the toolbar).

Step 5: Analyze Packets

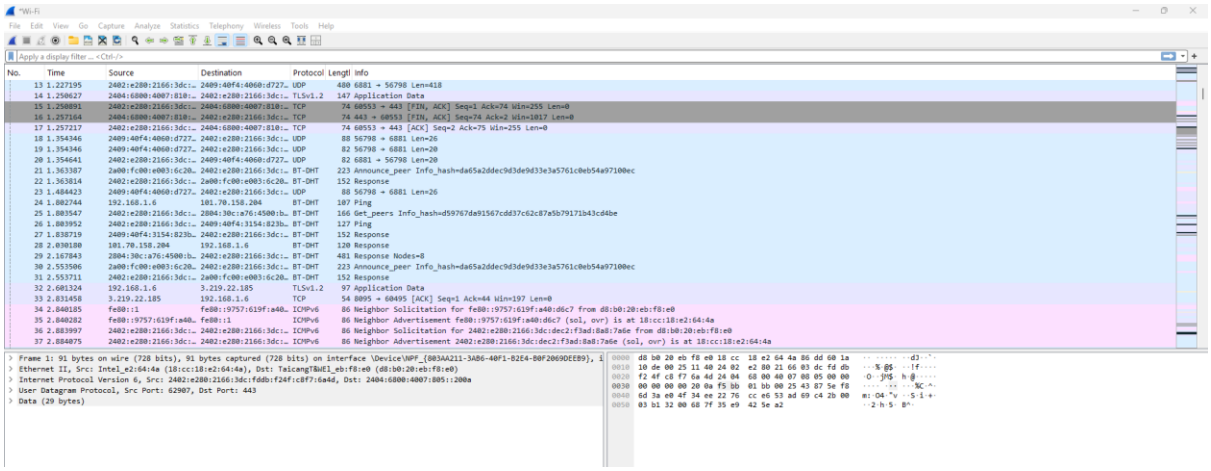
- Click on any packet to **expand its layers** (Ethernet, IP, TCP, etc.).
- View:

- Source & destination addresses
- Flags (like SYN, ACK in TCP)
- Payload data (especially useful for HTTP, DNS, FTP)

Step 6: Stop Capturing

- When done, click the **red square button (Stop)** in the toolbar.
- Optionally: Save the capture using File > Save As (.pcap format) for later analysis.

Output:



3a Configure a network topology using packet tracer software

Aim: To configure a basic network topology in Cisco Packet Tracer**,** which includes 2 PCs, 2 switches, and 1 router**, using** static IP addressing**.**

Network Topology Overview

PC0 --- SW0 --- R0 --- SW1 --- PC1

Step-by-Step Instructions

Step 1: Open Cisco Packet Tracer

- Launch Cisco Packet Tracer from your computer.
- Wait for the interface to fully load.

Step 2: Place Devices

From the bottom panel:

- Go to **End Devices** and drag:
  - 2 PCs to the workspace (PC0, PC1)
- Go to **Switches** and drag:
  - 2 switches (e.g., 2960) (Switch0, Switch1)
- Go to **Routers** and drag:
  - 1 router (e.g., Router0)

Step 3: Connect Devices Using Cables

Select the **"Connections" (lightning icon)** tool:

  - PC0 → Switch0 (FastEthernet0)
  - PC1 → Switch1 (FastEthernet0)
- Switch0 → Router0 (GigabitEthernet0/0)
- Router0 → Switch1 (GigabitEthernet0/1)

Make sure all links are green (active). If not, wait a few seconds.

Step 4: Configure IP Addresses

PC0:

- Click **PC0** → Desktop → IP Configuration:
  - IP Address: 192.168.1.10
  - Subnet Mask: 255.255.255.0
  - Default Gateway: 192.168.1.1

PC1:

- Click **PC1** → Desktop → IP Configuration:
  - IP Address: 192.168.2.10
  - Subnet Mask: 255.255.255.0

- Default Gateway: 192.168.2.1

Step 5: Configure the Router Interfaces

Click on **Router0** → CLI tab → Configure:

```
Router> enable
Router# configure terminal

# Configure GigabitEthernet0/0 (to Switch0 / PC0 side)
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

# Configure GigabitEthernet0/1 (to Switch1 / PC1 side)
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# exit
Router# write
```

Wait until both interfaces turn green (connected).

Step 6: Test the Network

1. Click **PC0** → Desktop → Command Prompt
   - Type: ping 192.168.2.10
2. Expected output: Successful replies (Reply from…)

This shows PC0 can reach PC1 through the router.

3b. To configure simple static routing.

AIM: To configure simple static routing in Cisco Packet Tracer**,** using two routers**,** two networks**,** and static routes**.** Configure Static Routing Between Two Networks

Network Diagram

[PC0]---[Switch0]---[Router0]---[Router1]---[Switch1]---[PC1]

Network A	Router Link	Network B
192.168.1.0/24	10.0.0.0/30	192.168.2.0/24

Step-by-Step Configuration

Step 1: Add Devices in Packet Tracer

- **2 PCs** (PC0, PC1)
- **2 Switches** (Switch0, Switch1)
- **2 Routers** (Router0, Router1)

Step 2: Connect Devices

- PC0 → Switch0
- PC1 → Switch1
- Switch0 → Router0 (e.g., G0/0)
- Switch1 → Router1 (e.g., G0/0)

Use **copper cross-over cable**:

- Router0 G0/1 → Router1 G0/1

PC0

- IP: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.1.1

PC1

- IP: 192.168.2.10
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.2.1

Step 4: Configure Router Interfaces

```
Router> enable
Router# configure terminal

# LAN interface
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

```
# Link to Router1
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit

# Add static route to 192.168.2.0/24 via Router1
Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2

Router(config)# exit
Router> enable
Router# configure terminal

# LAN interface
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

# Link to Router0
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 10.0.0.2 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit

# Add static route to 192.168.1.0/24 via Router0
Router(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1

Router(config)# exit
```

Step 5: Verify Routing

From **PC0**:

- Go to Desktop → Command Prompt
- Type: ping 192.168.2.10

From **PC1**:

- Type: ping 192.168.1.10

If routing is correct, you'll get **replies**.

Troubleshooting Tips

- Check interface status (show ip interface brief)
- Use ping between routers to test link
- Check static routes (show ip route)
- Make sure no shutdown was applied to interfaces

3C. To implement Security on interconnecting devices.

AIM: To implement basic security **on** interconnecting devices (routers and switches) using Cisco Packet Tracer.

Network Devices Involved

- **Router (R1)**
- **Switch (SW1)**

These are often the most vulnerable points in a network if not secured.

Step 3: Assign IP Addresses

Security Configuration Goals

Security Task	Devices Affected
Set passwords for console and VTY	Router, Switch
Encrypt passwords	Router, Switch
Set enable secret	Router, Switch
Disable unused switch ports	Switch only
Set a login banner	Router, Switch

Step-by-Step: Router Security (e.g., Router R1)

Step 1: Access CLI

Click on the router → **CLI Tab** → Press Enter

Step 2: Set Enable Password

```
Router> enable
Router# configure terminal
Router(config)# enable secret cisco123
```

This encrypts the enable password automatically.

Step 3: Set Console Line Password

```
Router(config)# line console 0
Router(config-line)# password conpass
Router(config-line)# login
Router(config-line)# exit
```

Step 4: Set VTY Line Password (for Telnet/SSH)

```
Router(config)# line vty 0 4
```

```
Router(config-line)# password vtypass
Router(config-line)# login
Router(config-line)# exit
```

Step 5: Encrypt Plain Text Passwords

```
Router(config)# service password-encryption
```

Step 6: Set a Security Banner

```
Router(config)# banner motd #Unauthorized Access is Prohibited!#
```

Step-by-Step: Switch Security (e.g., Switch SW1)

Follow similar steps for the switch:

Access CLI

Click on Switch → CLI Tab → Press Enter

Configure Enable Secret

```
Switch> enable
Switch# configure terminal
Switch(config)# enable secret cisco123
```

Console & VTY Passwords

```
Switch(config)# line console 0
Switch(config-line)# password conpass
Switch(config-line)# login
Switch(config-line)# exit

Switch(config)# line vty 0 4
Switch(config-line)# password vtypass
Switch(config-line)# login
Switch(config-line)# exit
```

Encrypt All Passwords

```
Switch(config)# service password-encryption
```

Set a Banner

```
Switch(config)# banner motd #Access restricted to authorized personnel only#
```

Step 7: Disable Unused Switch Ports (Best Practice)

Assuming FastEthernet 0/10 to 0/24 are unused:

```
Switch(config)# interface range fastEthernet 0/10 - 24
```

```
Switch(config-if-range)# shutdown
Switch(config-if-range)# exit
```

Verify Security Settings

After configuration, use the following commands:

- show running-config
- show line
- show interfaces status

3d. To configure network state routing protocol.

Aim: To configure a Link-State Routing Protocol (specifically OSPF**) in** Cisco Packet Tracer**,** follow this step-by-step guide. OSPF (Open Shortest Path First) is the most common link-state protocol used in enterprise networks. create a simple network with **2 routers**, **2 end devices**, and configure **OSPF** to allow them to communicate.

Network Topology
PC0 --- R1 --- R2 --- PC1

Network A: 192.168.1.0/24
Network B: 10.0.0.0/30 (between routers)
Network C: 192.168.2.0/24
Devices Used

- 2 PCs
- 2 Routers (e.g., Router0 and Router1)
- 2 Switches (optional for realism)

Step-by-Step Instructions

Step 1: Setup the Topology

1. Place the devices (PC0, R1, R2, PC1)
2. Connect them as follows using **straight-through cables**:
    - PC0 → Router1 (G0/0)
    - PC1 → Router2 (G0/0)
    - Router1 (G0/1) ↔ Router2 (G0/1) *(this is the OSPF link)*

Step 2: Assign IP Addresses

PC0:

- IP: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

PC1:

- IP: 192.168.2.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

Router1:
enable
configure terminal
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface g0/1
ip address 10.0.0.1 255.255.255.252

no shutdown
exit

Router2:
enable
configure terminal

interface g0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

interface g0/1
ip address 10.0.0.2 255.255.255.252
no shutdown
exit

Step 3: Configure OSPF (Link-State Routing Protocol)

On Router1:
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
exit
On Router2:
router ospf 1
network 192.168.2.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
exit

Step 4: Verify OSPF Neighbor Relationship

From either router:

>show ip ospf neighbor

Step 5: Test Connectivity

1. Go to **PC0** → Desktop → Command Prompt
2. Type: ping 192.168.2.10

If OSPF is working correctly, you'll get **replies**

Summary Table

Device	Interface	IP Address	Role
PC0	NIC	192.168.1.10	End host
Router1	G0/0	192.168.1.1	LAN side
Router1	G0/1	10.0.0.1	OSPF link

Device	Interface	IP Address	Role
Router2	G0/1	10.0.0.2	OSPF link
Router2	G0/0	192.168.2.1	LAN side
PC1	NIC	192.168.2.10	End host

3. CASE STUDY

3e. Study of Network simulator (NS).and Simulation of Congestion Control Algorithms using NS

Aim:
To Study of Network simulator (NS).and Simulation of Congestion Control Algorithms using NS

NET WORK SIMULATOR (NS2)

Ns overview

- Ns programming: A Quick start
- Case study I: A simple Wireless network
- Case study II: Create a new agent in Ns

Ns overview

- Ns Status
- Periodical release (ns-2.26, Feb 2003)
- Platform support
- FreeBSD, Linux, Solaris, Windows and Mac

Ns Functionalities

Routing, Transportation, Traffic sources, Queuing disciplines, QoS

Wireless

Ad hoc routing, mobile IP, sensor-MAC
Tracing, visualization and various utilitie

NS(Network Simulators)

Most of the commercial simulators are GUI driven, while some network simulators are CLI driven. The network model / configuration describes the state of the network (nodes,routers, switches, links) and the events (data transmissions, packet error etc.). An important output of simulations are the trace files. Trace files log every packet, every event that occurred in the simulation and are used for analysis. Network simulators can also provide other tools to facilitate visual analysis of trends and potential trouble spots.

Most network simulators use discrete event simulation, in which a list of pending "events" is stored, and those events are processed in order, with some events triggering future events—such as the event of the arrival of a packet at one node triggering the event of the arrival of that packet at a downstream node.

Simulation of networks is a very complex task. For example, if congestion is high, then estimation of the average occupancy is challenging because of high variance. To estimate the likelihood of a buffer overflow in a network, the time required for an accurate answer can be extremely large. Specialized techniques such as "control variates" and "importance sampling" have been developed to speed simulation.

Examples of network simulators

There are many both free/open-source and proprietary network simulators. Examples of notable network simulation software are, ordered after how often they are mentioned in research papers:

1. ns (open source)
2. OPNET (proprietary software)
3. NetSim (proprietary software)

Uses of network simulators

Network simulators serve a variety of needs. Compared to the cost and time involved in setting up an entire test bed containing multiple networked computers, routers and data links, network simulators are relatively fast and inexpensive. They allow engineers, researchers to test scenarios that might be particularly difficult or expensive to emulate using real hardware - for instance, simulating a scenario with several nodes or experimenting with a new protocol in the network. Network simulators are particularly useful in allowing researchers to test new networking protocols or changes to existing protocols in a controlled and reproducible environment. A typical network simulator encompasses a wide range of networking technologies and can help the users to build complex networks from basic building blocks such as a variety of nodes and links. With the help of simulators, one can design hierarchical networks using various types of nodes like computers, hubs, bridges, routers, switches, links, mobile units etc.

Various types of Wide Area Network (WAN) technologies like TCP, ATM, IP etc. and Local Area Network (LAN) technologies like Ethernet, token rings etc., can all be simulated with a typical simulator and the user can test, analyze various standard results apart from devising some novel protocol or strategy for routing etc. Network simulators are also widely used to simulate battlefield networks in Network-centric warfare

There are a wide variety of network simulators, ranging from the very simple to the very complex. Minimally, a network simulator must enable a user to represent a network topology, specifying the nodes on the network, the links between those nodes and the traffic between the nodes. More complicated systems may allow the user to specify everything about the protocols used to handle traffic in a network. Graphical applications allow users to easily visualize the workings of their simulated environment. Text-based applications may provide a less intuitive interface, but may permit more advanced forms of customization.

Packet loss

occurs when one or morepacketsof data travelling across a computer networkfail to reachtheir destination. Packet loss is distinguished as one of the three main error types encountered in digital communications; the other two being bit errorand spurious packets caused due to noise.

Packets can be lost in a network because they may be dropped when a queue in the network node overflows. The amount of packet loss during the steady state is another important property of a congestion control scheme. The larger the value of packet loss, the more difficult it is for transportlayer protocols to maintain high bandwidths, the sensitivity to loss of individual packets, as well as to frequency and patterns of loss among longer packet sequences is strongly dependent on the application itself.

Throughput

This is the main performance measure characteristic, and most widely used. Incommunicationnetworks, such asEthernetorpacket radio, throughputor network throughputis the average rate of successfulmessage delivery over a communication channel. The throughput is usually measured inbitsper second (bit/s orbps), andsometimes indata packetsper second or data packets pertime slotThis measure how soon the receiver is able to get a certain amount of data send by the sender. It is determined as the ratio of the total data received to the end to end delay. Throughput is an important factor which directly impacts the network performance

Delay

Delay is the time elapsed while a packet travels from one point e.g., source premise or network ingress to destination premise or network degrees. The larger the valueof delay, the more difficult it is for transport layer protocols to maintain highbandwidths. We will calculate end to end delay

2. Case study: Obtaining the MAC Address of a Device Using ARP in Cisco Packet Tracer

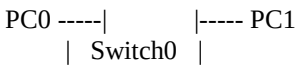
Aim:

To simulate a network in Cisco Packet Tracer and use the **Address Resolution Protocol (ARP)** to retrieve the **MAC address** of a connected device.

Requirements:

- Cisco Packet Tracer (any recent version)
- 2 or more PCs connected via a switch or router
- IP address configuration completed on both PCs

Network Topology:



Procedure:

Step 1: Setup the Network

1. **Open Cisco Packet Tracer**
2. **Drag and drop**:
    - 2 **PCs** (PC0 and PC1)
    - 1 **Switch**
    - Connect them using **copper straight-through cables**
3. Use **"FastEthernet"** ports to connect:
    - PC0 → Switch (Fa0/1)
    - PC1 → Switch (Fa0/2)

Step 2: Assign IP Addresses

1. Click on **PC0** → Desktop tab → IP Configuration:
    - IP: `192.168.1.1`
    - Subnet Mask: `255.255.255.0 (automatically assigned)`
2. Click on **PC1** → Desktop tab → IP Configuration:
    - IP: `192.168.1.2`
    - Subnet Mask: `255.255.255.0 (automatically assigned)`

Step 3: Ping the Target to Generate ARP Entry

1. Go to **PC0** → Desktop tab → Command Prompt
2. Type the command:

   ```
   ping 192.168.1.2
   ```

3. This generates ARP traffic and resolves the MAC of PC1.

Step 4: Use ARP Command to View MAC Address

1. On the same PC0 command prompt, type:

   ```
   arp -a
   ```

   It shows the IP and the corresponding MAC (Physical) Address of PC1.

Output :

```
Interface: 192.168.1.1 --- 0x1
  Internet Address      Physical Address      Type
  192.168.1.2           00-0c-29-aa-bb-cc     dynamic
```

Result:

The MAC address of PC1 (`192.168.1.2`) was successfully retrieved using the ARP command on PC0.